



LightCOM

Lighthouse Consortium on
Obesity Management

Study name: **Lighthouse Consortium on Obesity Management**

Short name: **LightCOM**

SOP title: **Standard Operating Procedure for the measures of
body weight, height, and waist circumference**

SOP short title: **SOP on Anthropometry**

SOP Version: **1**

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TABLE OF CONTENTS

1	PURPOSE AND APPLICABILITY	3
2	PERSONNEL QUALIFICATIONS AND RESPONSIBILITIES	3
3	EQUIPMENT AND RESOURCES.....	3
3.1	Equipment	3
3.2	Forms	3
3.3	Other Resources	3
4	HEALTH AND SAFETY	3
5	PROCEDURE	3
5.1	Measurement of weight	4
5.2	Measurement of height.....	4
6	EXPECTED RESULTS, TROUBLESHOOTING AND CORRECTIVE ACTION	5
7	DATA AND RECORDS MANAGEMENT.....	5
8	REFERENCE.....	6
9	HISTORICAL RECORD of SOP	6

1 PURPOSE AND APPLICABILITY

This document outlines the procedure to follow whilst undertaking measures of height, weight, and waist circumference from participants of the LightCOM trial. The document complies with procedures described in the WHO STEPS guideline for physical measurements.

2 PERSONNEL QUALIFICATIONS AND RESPONSIBILITIES

- Weight measurement training
- Height measure training
- Waist circumference measurement training

This SOP should be referred to for training purposes. All practical training should be conducted by an experienced staff member.

3 EQUIPMENT AND RESOURCES

3.1 Equipment

- *Weight*: digital weighing scales
- *Height*: stadiometer
- *Waist*: constant tension tape (for example, Figure Finder Tape Measure) and pen

3.2 Forms

- Training checklist and staff training log
- Participant eCRF in OpenClinica

3.3 Other Resources

- NA

4 HEALTH AND SAFETY

- NA

5 PROCEDURE

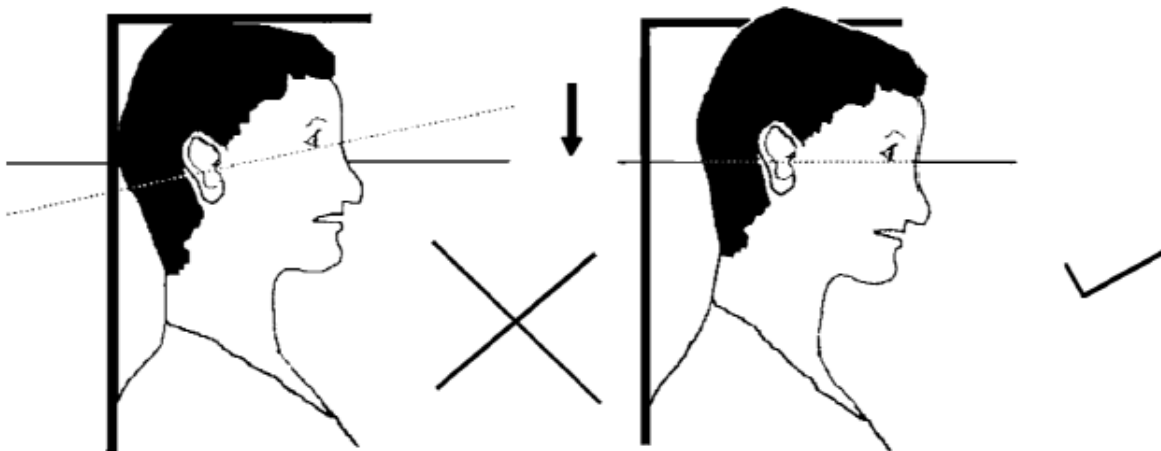
- Measurements will be taken by a trained member of the research team
- The participant will be wearing light clothing at all assessments

5.1 Measurement of weight

- Prepare the scale in advance, according to instructions.
- Participants should remove all heavy clothing and shoes before stepping onto the scale (so that the measurement is as accurate as possible).
- Participants should stand in a natural position on the scale with their feet centred on the scale. Arms should hang freely by sides with head facing forward and looking ahead, as posture is important.
- Follow the instructions for use according to the user manual.
- Record reading in kg, to one decimal point (e.g. 64.3 kg) before the participant steps off the scale. Ensure that the reading is recorded in the eCRF.

5.2 Measurement of height

- Assemble the stadiometer in advance, according to instructions.
- Participants should remove their shoes before stepping onto the stadiometer (so measurement is as accurate as possible).
- Participants should stand facing forward (with vertical rod at back, feet flat and together on centre of base plate, weight evenly distributed). Arms should hang freely by sides and head, back, buttocks and heels against vertical rod (but not leaning on it).
- Ensure participant does not lift their heels off the base plate.
- With both hands, position the participant's head in the Frankfurt Plane* and lower the head-plate firmly on the head.
- Record measurement in cm (to the nearest 0.1 cm) in the eCRF.
- Instruct participant to step away from the stadiometer.



*The Frankfurt plane is an imaginary horizontal line passing through the external ear canal and across the top of the lower bone of the eye socket, immediately under the eye. This position is important if an accurate reading is to be obtained. If participant is tall, it can be difficult to line up the Frankfurt Plane in the way described. When you think that the plane is horizontal, take one step back to check from a short distance that this is the case.

5.3 Measurement of waist circumference

- DK only: Provide the participant with a plain white t-shirt, or similar, to wear during clinical assessments (LightBAR only).
- Participants should also remove their belt and they may need to unbutton the upper trouser button.
- Participants should stand upright in a relaxed manner, breathing normally with arms hanging loosely by sides. Researchers should tell participants to breathe in and out naturally and not hold their breath, so the measurement is as accurate as possible. Their weight should be evenly balanced on both feet.
- The waist measurement should be taken at the midpoint between the lower margin of the last palpable rib and the top of the iliac crest (hip bone):
 - *Step 1:* Standing to the side of the participant, locate and mark the last palpable rib and the top of the hip bone. Use the heel of your hand in a rolling motion to help locate the top of the hip bone, this will increase the surface area you are feeling as well as being more comfortable for the participant. It can be difficult to locate these points, especially on individuals with obesity. Ask the participant to assist you in locating these points on their body if needed.
 - *Step 2:* With a tape measure, find the midpoint and mark the point (either with a pen mark if on skin, or by placing a finger on the point if wearing a t-shirt).
 - *Step 3:* Ask the participant to wrap the tension tape around themselves and then position the tape at the midpoint of the last palpable rib and the top of the hip bone, making sure to wrap the tape over the same spot on the opposite side.
Note: Check that the tape is horizontal across the back and front of the participant and as parallel to the floor as possible.
 - *Step 4:* Ask the participant to breathe in and breathe out gently and to look straight ahead (to stop them contracting their muscles or holding their breath).
 - *Step 5:* Record measurement after breathing out, at the level of the tape to the nearest 0.1 cm, making sure to keep the measuring tape snug but not tight enough to cause compression of the skin.
 - *Step 6:* Repeat steps 3-5. If there more than a 1 cm difference between the first and second measurements do a third measure.
 - *Step 7:* Record the measurements on the participant's eCRF.

6 EXPECTED RESULTS, TROUBLESHOOTING AND CORRECTIVE ACTION

- NA

7 DATA AND RECORDS MANAGEMENT

- All data will be recorded in OpenClinica at all CIDs.

8 REFERENCE

- World Health Organisation (2008). Guide to Physical Measurements. WHO Steps Surveillance.

9 HISTORICAL RECORD of SOP

Rev# / Date	Written/Revised By	Revision Description	Approval
0. Written	Sofus C. Larsen		
1. Edited			
2.			
3.			
4.			
5.			
6.			
Removed / replaced:			